Unit 6: Dimensionality Reduction

Curse of dimensionality is the problem that happens when the number of **features (dimensions)** in a dataset becomes very large.

As dimensions increase, **data points become sparse**, and many machine learning methods become **less effective, slower, and less accurate**.

Simple Meaning

Imagine:

- In **1D**, data points are on a line.
- In **2D**, they are on a plane.
- In **3D**, they are in space.

Now imagine **100 dimensions**.
The space becomes so huge that the available data points are **far apart** from each other.

So even if you have “a lot” of data, it may still be **too little** for such a high-dimensional space.

Why is it called a “curse”?

Because increasing dimensions creates many difficulties:

1) Data becomes sparse

In high dimensions, points are spread out.

➞ This means:

- Hard to find similar data points
- Harder to form clusters
- Harder to generalize patterns

2) Distance measures become less meaningful

Algorithms like:

- **KNN**
- **K-means clustering**

- **Hierarchical clustering**

depend on distance.

But in high dimensions:

- The **nearest point** and **farthest point** may become almost equally distant.
- So “closeness” loses meaning.

➞ Distance-based algorithms perform poorly.

3) Need much more data

More features require more training examples.

For example:

- 2 features → manageable data requirement
- 100 features → enormous data requirement

➞ If you don’t have enough samples, the model may **overfit**.

4) Computation becomes expensive

High-dimensional data increases:

- memory usage
- training time
- model complexity

➞ The algorithm becomes slower and harder to optimize.

5) Visualization becomes difficult

You can visualize:

- 2D data easily
- 3D data somewhat

But beyond that?

➞ Human understanding becomes harder.

Example

Suppose you are building a customer prediction model.

Case 1: 2 features

- Age
- Income

You can plot and analyze the data easily.

Case 2: 100 features

Now you include:

- Age
- Income
- purchase frequency
- screen time
- clicks
- region
- app behavior
- browsing history
- ...and 93 more

Now:

- many features may be irrelevant
- points become sparse
- model may overfit
- training becomes harder

This is the **curse of dimensionality**.

Feature Selection vs Feature Extraction

Both are used to **reduce dimensionality** and improve model performance, but they work differently.

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1) What is a Feature?

A **feature** is an input variable used to train a machine learning model.

Example:

If you are predicting student performance, features can be:

- Study hours
- Attendance
- Internal marks
- Sleep hours

These are the inputs to the model.

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2) Feature Selection

Definition

Feature Selection means **choosing the most important features from the original dataset** and removing unnecessary ones.

👉 It does **not create new features**.
👉 It only **selects useful existing features**.

Example

Suppose a dataset has 6 features:

- Age
- Height
- Weight
- Name

- Roll Number
- Marks

For predicting health risk, useful features may be:

- Age
- Height
- Weight

Unnecessary features may be:

- Name
- Roll Number

So feature selection keeps only relevant features.

Result:

From:

[Age, Height, Weight, Name, Roll Number, Marks]

To:

[Age, Height, Weight]

Why Feature Selection is Used

It helps to:

- reduce irrelevant data
- improve model accuracy
- reduce overfitting
- decrease training time
- make model easier to interpret

Types of Feature Selection

A) Filter Methods

Features are selected using statistical tests.

Examples:

- Correlation
- Chi-square test
- ANOVA
- Mutual Information

Example:

If two features are highly correlated, one may be removed.

B) Wrapper Methods

Different combinations of features are tested using a model.

Examples:

- Forward Selection
- Backward Elimination
- Recursive Feature Elimination (RFE)

C) Embedded Methods

Feature selection happens during model training.

Examples:

- Lasso Regression (L1)
- Decision Trees
- Random Forest

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3) Feature Extraction

Definition

Feature Extraction means **transforming the original features into a new set of features**.

👉 It **creates new features**
👉 These new features contain the most useful information from the original data

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Example

Suppose you have 5 correlated features:

- Math marks
- Physics marks
- Chemistry marks
- Biology marks
- English marks

Instead of using all separately, feature extraction may combine them into:

- Academic Performance Score
- Science Ability Score

These are **new features** made from the old ones.

Why Feature Extraction is Used

It helps to:

- reduce dimensionality
- remove redundancy
- compress data
- improve model performance
- make complex data easier to process

Common Feature Extraction Techniques

A) PCA (Principal Component Analysis)

Converts original features into **principal components**.

Example:
100 features → 10 principal components

B) LDA (Linear Discriminant Analysis)

Creates new features that help separate classes.

Used in classification problems.

C) Autoencoders

A deep learning method used to learn compressed features.

D) t-SNE / UMAP

Used mainly for visualization and representation of high-dimensional data.

Key Difference

Basis	**Feature Selection**	**Feature Extraction**
Meaning	Selects important existing features	Creates new features from old ones
Original features	Retained	Transformed
New features created?	No	Yes
Interpretability	Easy to understand	Harder to interpret
Example	Choose Age, Salary, Experience	Convert many variables into principal components
Main goal	Remove irrelevant features	Compress and transform data

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Simple Analogy

Feature Selection

Imagine you have **10 books** but only **3 are useful** for exam preparation.

➡ You **select** only those 3 books.

That is **Feature Selection**.

Feature Extraction

Now imagine you read all 10 books and prepare **short notes** from them.

➡ You create **new summarized material**.

That is **Feature Extraction**.

Real-Life Example in ML

Problem:

Predict whether a customer will buy a product.

Original features:

- Age
- Salary
- City
- Number of clicks
- Time on website
- Browser type
- Device type
- Last purchase amount

Using Feature Selection

You may keep only:

- Salary

- Number of clicks
- Time on website
- Last purchase amount

Removed:

- Browser type
- City (if not useful)

Using Feature Extraction

You may create new features like:

- Engagement Score
- Purchasing Power Score

These are combinations of original features.

Principal Component Analysis (PCA) is a **dimensionality reduction** technique used to reduce the number of features in a dataset while preserving as much information (variance) as possible.

In simple words:

PCA converts many original features into a smaller number of new features called Principal Components.

Simple Definition

PCA is a statistical technique that transforms correlated variables into a smaller set of uncorrelated variables called principal components.

Why PCA is Used

When a dataset has:

- too many features
- highly correlated variables
- unnecessary complexity

PCA helps by:

- reducing dimensionality
- removing redundancy
- improving model efficiency
- helping in visualization
- reducing overfitting

Basic Idea of PCA

Suppose a dataset has 100 features.

Instead of using all 100 features, PCA may convert them into:

- PC1
- PC2
- PC3
- ...

- PC10

These **10 principal components** capture most of the important information from the original 100 features.

What is a Principal Component?

A **Principal Component** is a **new variable** formed by combining original variables.

Important properties:

- Principal components are **linear combinations** of original features
- They are **uncorrelated**
- They are arranged in order of importance

Order of Principal Components

1) First Principal Component (PC1)

- Captures the **maximum variance** in the data

2) Second Principal Component (PC2)

- Captures the **next highest variance**
- Perpendicular (orthogonal) to PC1

3) Third Principal Component (PC3)

- Captures the next variance

...and so on.

Example

Suppose student data has these correlated features:

- Math marks
- Physics marks
- Chemistry marks

These subjects may be strongly related.

PCA can combine them into:

- **PC1 = Science Performance**
- **PC2 = Subject Variation**

So instead of 3 variables, we may use 2 principal components.

PCA tries to find the **directions in which the data varies the most**.

Think of it like this:

If data points are spread diagonally on a graph, PCA rotates the axis so that the new axis captures the spread better.

Steps in PCA

Step 1: Standardize the data

Since features may have different scales, we first normalize/standardize them.

Example:

- Age: 18–60
- Income: 10,000–1,00,000

Without scaling, larger values dominate.

Step 2: Compute covariance matrix

Covariance shows how features vary together.

If two features are highly correlated, PCA can combine them.

Step 3: Find eigenvalues and eigenvectors

This is the mathematical core of PCA.

- **Eigenvectors** → directions of principal components
- **Eigenvalues** → amount of variance captured

Step 4: Sort components

Components are ranked based on eigenvalues.

Highest eigenvalue = most important component

Step 5: Select top components

Choose the first few principal components that explain most of the variance.

Example:

- 100 features → top 10 components

Step 6: Transform the data

Original data is projected onto the new principal component axes.

Example with Features

Suppose a customer dataset has:

- Annual Income
- Spending Score
- Purchase Frequency
- Website Visits
- Time on App

PCA may convert them into:

- **PC1 = Customer Engagement**
- **PC2 = Spending Power**

This reduces complexity.

Advantages of PCA

1) Reduces dimensionality

Less number of features

2) Removes multicollinearity

Correlated features are transformed into uncorrelated components

3) Speeds up training

Fewer features = faster model training

4) Helps visualization

High-dimensional data can be reduced to 2D or 3D for plotting

5) Reduces overfitting

By removing noise and redundant data.

Where PCA is Used

PCA is used in:

- Machine Learning preprocessing
- Image compression
- Face recognition
- Bioinformatics
- Finance
- Pattern recognition
- Data visualization

Difference Between Feature Selection and PCA

Feature Selection	**PCA**
Chooses important existing features	Creates new transformed features
Original features remain	Original features are transformed
Easy to interpret	Harder to interpret

If original variables are:

- X_1
- X_2
- X_3

Then a principal component may look like:

$PC1 = a_1X_1 + a_2X_2 + a_3X_3$

where:

- **a_1, a_2, a_3** are weights

Example: Covariance Matrix Calculation

Suppose we have a small dataset with **2 features**:

- **X = Study Hours**
- **Y = Exam Marks**

Dataset

Student	X (Study Hours)	Y (Marks)
1	2	50
2	4	60
3	6	65
4	8	80

We will calculate the **covariance matrix** for X and Y.

Step 1: Find Mean of Each Feature

Mean of X

$$\bar{X} = \frac{2+4+6+8}{4} = \frac{20}{4} = 5$$

Mean of Y

$$\bar{Y} = \frac{50+60+65+80}{4} = \frac{255}{4} = 63.75$$

Step 2: Find Deviations from Mean

Student	X	Y	$X-\bar{X}$	$Y-\bar{Y}$
1	2	50	-3	-13.75
2	4	60	-1	-3.75
3	6	65	1	1.25
4	8	80	3	16.25

Step 3: Calculate Variance and Covariance Terms

We need:

- **Var(X)**
- **Var(Y)**
- **Cov(X,Y)**

Using sample covariance formula:

$$\text{Cov}(X,Y) = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$

where $n = 4$

A) Calculate Variance of X

$$\begin{aligned}\text{Var}(X) &= \frac{\sum(X_i - \bar{X})^2}{n-1} \\ &= \frac{(-3)^2 + (-1)^2 + (1)^2 + (3)^2}{3} \\ &= \frac{9+1+1+9}{3} \\ &= \frac{20}{3} = 6.67\end{aligned}$$

B) Calculate Variance of Y

$$\begin{aligned}\text{Var}(Y) &= \frac{\sum(Y_i - \bar{Y})^2}{n-1} \\ &= \frac{(-13.75)^2 + (-3.75)^2 + (1.25)^2 + (16.25)^2}{3} \\ &= \frac{189.06 + 14.06 + 1.56 + 264.06}{3} \\ &= \frac{468.74}{3} = 156.25\end{aligned}$$

C) Calculate Covariance between X and Y

$$\text{Cov}(X,Y) = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$

Now calculate products:

Student	$X - \bar{X}$	$Y - \bar{Y}$	**Product**
1	-3	-13.75	41.25
2	-1	-3.75	3.75
3	1	1.25	1.25
4	3	16.25	48.75

$$\sum(X_i - \bar{X})(Y_i - \bar{Y}) = 41.25 + 3.75 + 1.25 + 48.75 = 95$$
$$\text{Cov}(X,Y) = \frac{95}{3} = 31.67$$

Step 4: Form the Covariance Matrix

The covariance matrix is:

$$\text{Cov Matrix} = \begin{bmatrix} \text{Var}(X) & \text{Cov}(X,Y) \\ \text{Cov}(Y,X) & \text{Var}(Y) \end{bmatrix}$$

Substitute values:

$$\text{Cov Matrix} = \begin{bmatrix} 6.67 & 31.67 \\ 31.67 & 156.25 \end{bmatrix}$$

Final Answer

Covariance Matrix

$$\boxed{\begin{bmatrix} 6.67 & 31.67 \\ 31.67 & 156.25 \end{bmatrix}}$$

Interpretation

1) Variance of X = 6.67

Study hours vary moderately.

2) Variance of Y = 156.25

Marks have larger spread.

3) Covariance = 31.67

This is **positive**, so:

➞ When **study hours increase**, **marks also tend to increase**.

That means X and Y are **positively related**.

If covariance is:

- **Positive** → both increase together
- **Negative** → one increases, the other decreases
- **Zero / near zero** → no linear relationship

Why Covariance Matrix is Important in PCA

In **PCA**, the covariance matrix helps us understand:

- which features are correlated
- how data varies together
- which directions contain maximum variance

Then PCA finds **eigenvalues and eigenvectors** from this matrix.